i-stop CONTROL [SKYACTIV-D 1.5]

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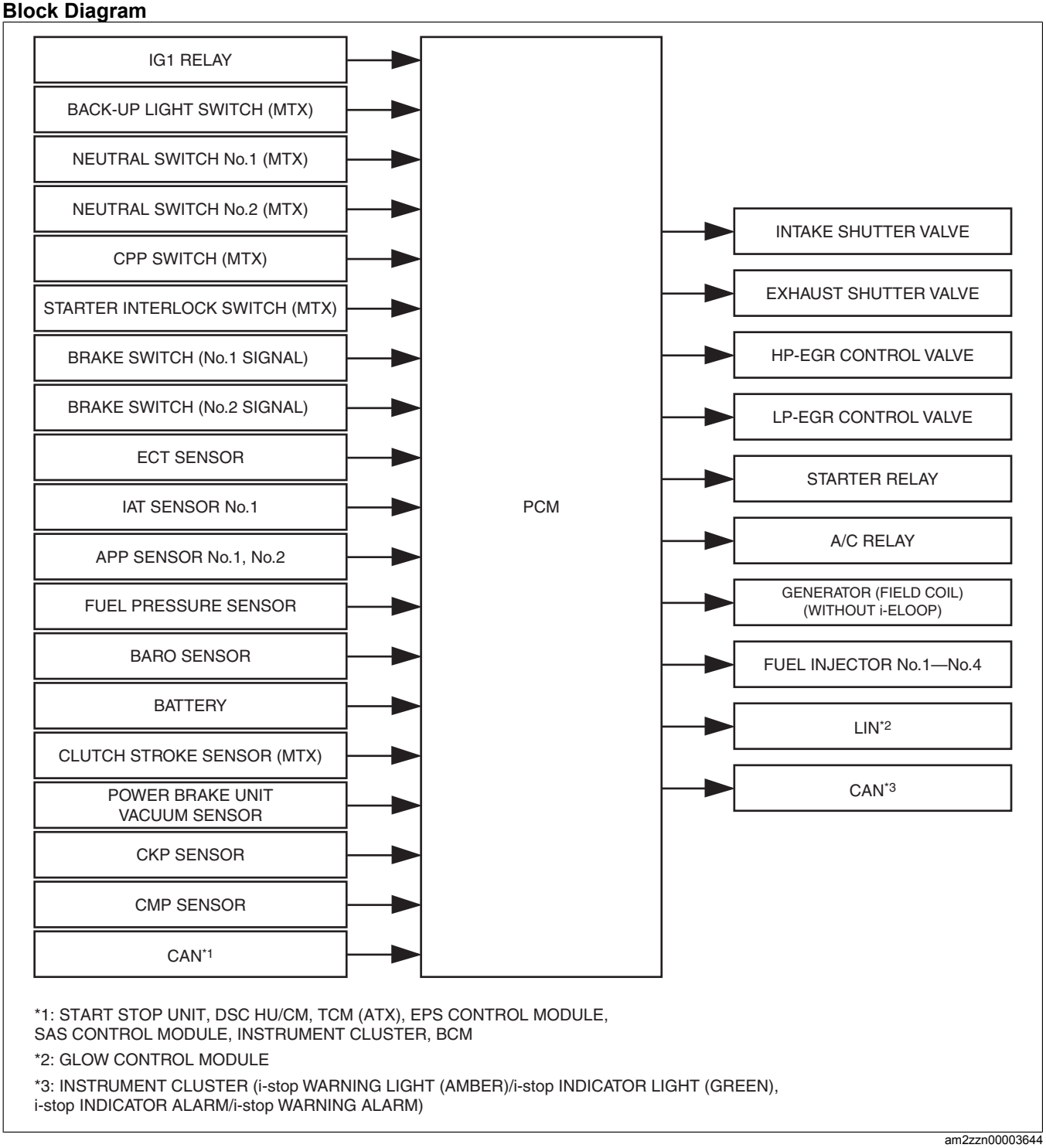
Outline

- When the vehicle is stopped such as at a stop light, the i-stop control stops/starts the engine automatically to improve fuel economy and reduce exhaust gas and idling noise.
- The PCM determines whether to permit/inhibit i-stop control based on the signal from each input part and CAN communication.
- The i-stop control includes the engine stop control, engine restart control, electric AT oil pump driver control, and hill launch assist functions.

Control Table

Control name	Control outline
Engine stop control	• The engine is stopped when the i-stop (engine stop control) conditions are met. (See Engine stop control.)
Engine restart control	• The engine restarts when the i-stop (engine restart control) conditions are met. (See Engine restart control .)
Electric AT oil pump driver control (ATX)	• Drives the electric AT oil pump to assure line pressure during an engine stop by the i-stop control. (See ELECTRIC AT OIL PUMP [EW6A-EL].)
Vehicle roll prevention function (ATX)	• Controls the traction control solenoid valve in the DSC HU/CM to maintain or decrease brake fluid pressure. (See VEHICLE ROLL PREVENTION FUNCTION.)

Block Diagram



*1: START STOP UNIT, DSC HU/CM, TCM (ATX), EPS CONTROL MODULE, SAS CONTROL MODULE, INSTRUMENT CLUSTER, BCM

*2: GLOW CONTROL MODULE

*3: INSTRUMENT CLUSTER (i-stop WARNING LIGHT (AMBER)/i-stop INDICATOR LIGHT (GREEN), i-stop INDICATOR ALARM/i-stop WARNING ALARM)

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Operation

Engine stop control

i-stop (engine-stop control) permit condition

- The conditions to stop the engine by the i-stop control are as follows:

Determination module	Determination item	Determination condition		Applicable PID
		ATX	MTX	
PCM	Vehicle speed	0 km/h {0 mph}	3 km/h {2 mph} or less	WSPD_SEN_LF
				WSPD_SEN_RF
				WSPD_SEN_LR
				WSPD_SEN_RR
	Brake pedal	Brake pedal is depressed while in D or M position	Not applicable	TR
				BOO, BPA
	Brake fluid pressure	Brake fluid pressure is 1.7 MPa {17 kgf/cm^2, 247 psi} or more while in D or M position (perform correction according to vehicle inclination angle)	Not applicable	TR
				BRK_F_P_C
	Accelerator pedal	Released	←	APP
	Clutch pedal	Not applicable	Released	CPP
	MTX gear position	Not applicable	Neutral	CPP/PNP
	Vehicle conditions	Vehicle is stopped in D or M position*1, *8	Not applicable	TR
	Steering speed	15 deg/s or less	←	STR_ROT_SPD
	Steering angle	Within 65° of center to left and right (After EPS control module learns center value)	Not applicable	STR_AB_EST
	Steering torque	1.1 N·m {11 kgf·cm, 9.7 in·lbf} or less	←	STR_TRQ_S_M
	i-stop OFF switch	OFF	←	I-Stop_OFF
	Vehicle speed history (After engine started and driven once)	3 km/h {2 mph} or more	4 km/h {2.5 mph} or more	I-Stop_VSP
	i-ELOOP bypass mode operation condition	Bypass mode is not operated (BYPS_STAT: Bypass Off)	←	BYPS_STAT
	Capacitor (i-ELOOP) voltage	14 V or more	←	DCDC_IN_V
	Battery charge condition	65.7 % or more	←	BATT_SOC
	Battery fluid temperature	0—70 °C {32—158 °F}	←	BATT_TEMP
	Battery voltage	11.2 V or more	←	BATT_V
	Power brake unit vacuum	−45 kPa {−0.46 kgf/cm^2, −6.5 psi} or less (value calculated by subtracting BARO from BBP)	←	BBP
				BARO
	Door (driver's side)	Closed*3	←	DOOR_D
	Bonnet	Closed*3	←	HOOD
PCM	Vehicle inclination angle	Within ±7° when level (LON_ACCL_C: within ±0.121 G)	Not applicable	LON_ACCL_C
	Seat belt (driver's side)	Fastened *4	←	SEAT_B_D
	Fuel injection amount learning	Not implemented*5	←	Not applicable
	Diesel particulate filter regeneration	Not implemented	←	Not applicable
	Battery condition setting	Completed	←	Not applicable
	Steering angle sensor initial setting	Completed	Not applicable	Not applicable
	Engine coolant temperature	30—110 °C {86—230 °F}	←	ECT
	Intake air temperature	100 °C {212 °F} or less	←	IAT
	Altitude	1,800 m or less (BARO: 81.5 kPa {0.831 kgf/cm^2, 11.8 psi} or more)	←	BARO
	System condition	i-stop related part DTCs are not recorded	←	Not applicable

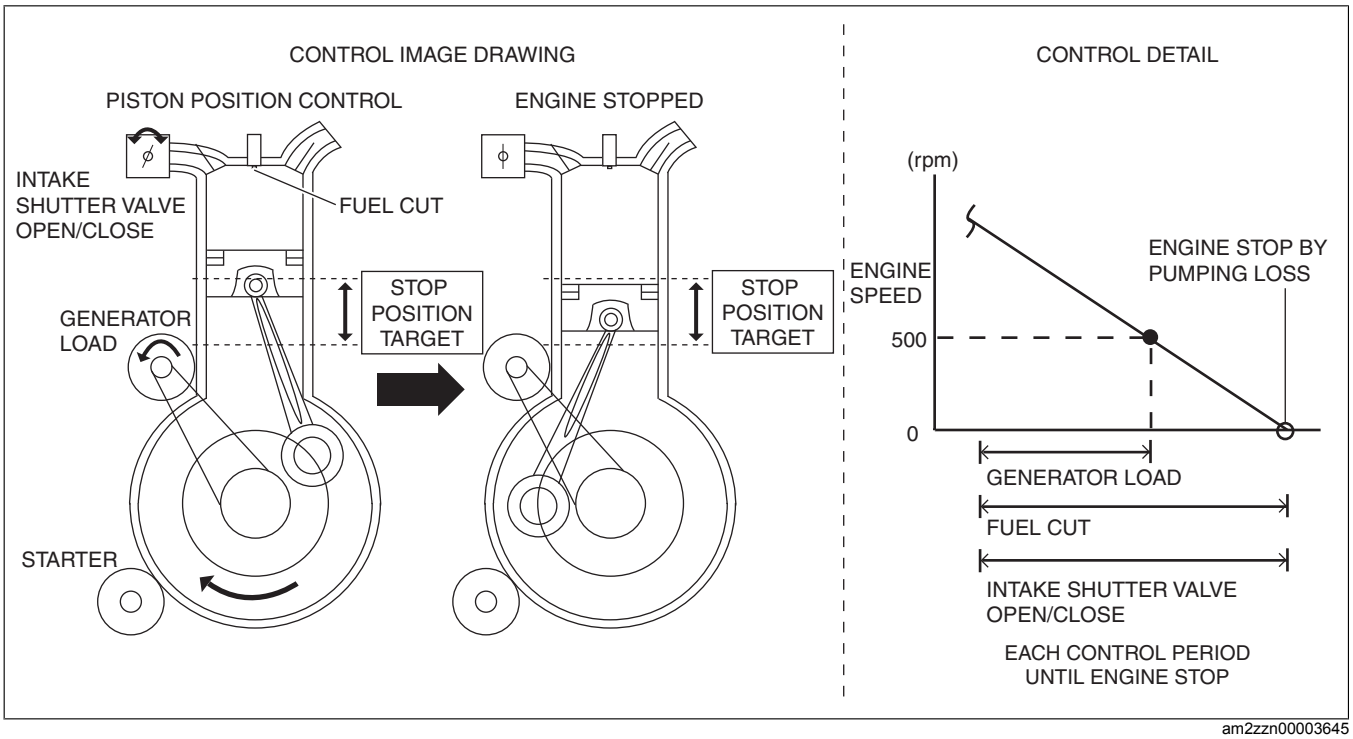
Determination module	Determination item	Determination condition		Applicable PID
		ATX	MTX	
Start stop unit	+B (ROOM fuse) input	Yes	←	VPWR_B1
	IG1 input	Yes	←	VPWR_IG1
	ACC input	Yes	←	VPWR_ACC
	IG1 output	Yes	←	RELAY_IG1
	ACC output	Yes	←	RELAY_ACC
	Push button	Not pressed	←	SSB_1
				SSB_2
	Door (all)	Closed	←	DOOR_ALL
	Trunk (4SD)/Liftgate (5HB)	Closed	←	TRUNK
	Seat belt (driver's side)	Fastened *4	←	SEAT_B_D
	Engine start	Engine was started by operating push button with key inside cabin (engine started first time)	←	Not applicable
	Engine rotation speed	300 rpm or more is maintained after engine started	←	Not applicable
	System condition	i-stop related part DTCs are not recorded	←	Not applicable
TCM	TCM detection condition	All met (I-Stop_EST: On)	Not applicable	I-Stop_EST
	Panic stop	Brake pedal was not depressed suddenly within 4 s	Not applicable	Not applicable
	TFT	20—117 °C {68—242 °F}	Not applicable	TFT
	Gear position	Not in 2nd gear fixed mode	Not applicable	Not applicable
	System condition	i-stop related part DTCs are not recorded	Not applicable	Not applicable
Climate control unit	Cabin temperature (with full-auto air conditioner system)	Difference between target cabin temperature and actual cabin temperature is a certain value or less (Cabin temperature control by A/C is normally performed)	←	Not applicable
	A/C temperature (with full-auto air conditioner system)	Setting other than MIN *6	←	Not applicable
	Ambient temperature	−10—50 °C {14—122 °F}	←	Not applicable
	Warm-up operation condition (with manual air conditioner system)	Ambient temperature is 10 °C {50 °F} or more or engine coolant temperature is 60 °C {140 ° F} or more	←	ECT
	Cold condition (with manual air conditioner system)	Ambient temperature is 20 °C {68 °F} or less or evaporator temperature is specified value or less	←	Not applicable
	Cold condition (with full-auto air conditioner system)	Ambient temperature is 4 ° C {39 °F} or more or evaporator temperature is 12 °C {54 °F} or less	←	Not applicable
	MODE dial	Except DEFROSTER	←	Not applicable
	Airflow volume control dial (with full-auto air conditioner system)	OFF *7	←	Not applicable
	System condition	i-stop related part DTCs are not recorded	←	Not applicable

Determination module	Determination item	Determination condition		Applicable PID
		ATX	MTX	
DSC HU/CM	Operation request from PCM	Hill Launch Assist (HLA) control request signal from PCM is received	Not applicable	Not applicable
	Wheel speed (4 wheels)	0 km/h {0 mph}	Not applicable	WSPD_SEN_LF
				WSPD_SEN_RF
				WSPD_SEN_LR
				WSPD_SEN_RR
	Brake pedal	Brake pedal is depressed	Not applicable	BRAKE_SW
	Brake fluid pressure	Approx. 0.45 MPa {4.6 kgf/cm^2, 65 psi} or more	Not applicable	BRK_F_P_C
	Accelerator pedal	Released	Not applicable	APP
	System condition	i-stop related part DTCs are not recorded	Not applicable	Not applicable

*1 : After the vehicle is stopped, the engine stops if the gear is shifted to N position. In addition, if the gear is shifted to P position after the vehicle is stopped in D position, the engine stop condition by the i-stop control is maintained.
*2 : If a door is opened while the vehicle is stopped, i-stop is inhibited until a vehicle speed record is input once..
*3 : If the engine is started with the bonnet open, i-stop is inhibited until the engine is stopped.
*4 : If the seat belt is unfastened while the vehicle is stopped, i-stop is inhibited until a vehicle speed record is input once.
*5 : If the fuel injection amount learning, which is periodically performed automatically, is not completed, the PCM turns the i-stop indicator light (green) off and inhibits i-stop when the vehicle speed is approx. 10 km/h {6.2 mph} or less before the vehicle stops.
*6 : When the A/C set temperature is set to MAX, the PCM may inhibit the i-stop control.
*7 : When all of the other i-stop permit conditions related to the climate control unit are met even if the airflow volume control dial is on, the i-stop control is permitted.
*8 : When shifting to the R position and driving the vehicle, the i-stop function is inhibited until the vehicle is driven at 5 km/h {3.0 mph} or more for 1 min or more after shifting to the D or M position.

i-stop (engine stop control)

- When the i-stop (engine stop control) permit conditions are met, the PCM stops the engine based on the following controls:
  - — Fuel injection control (fuel cut)
    - Engine speed is reduced by the fuel cut.
  - — Intake shutter valve control (intake shutter valve open/closed)
    - By adjusting the intake shutter opening angle, the engine speed is reduced and the pumping loss is used to stop the engine.
  - — Generator output control (generator load)
    - Engine speed is reduced by lowering the generator load.



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Engine restart control

i-stop (engine restart control) conditions

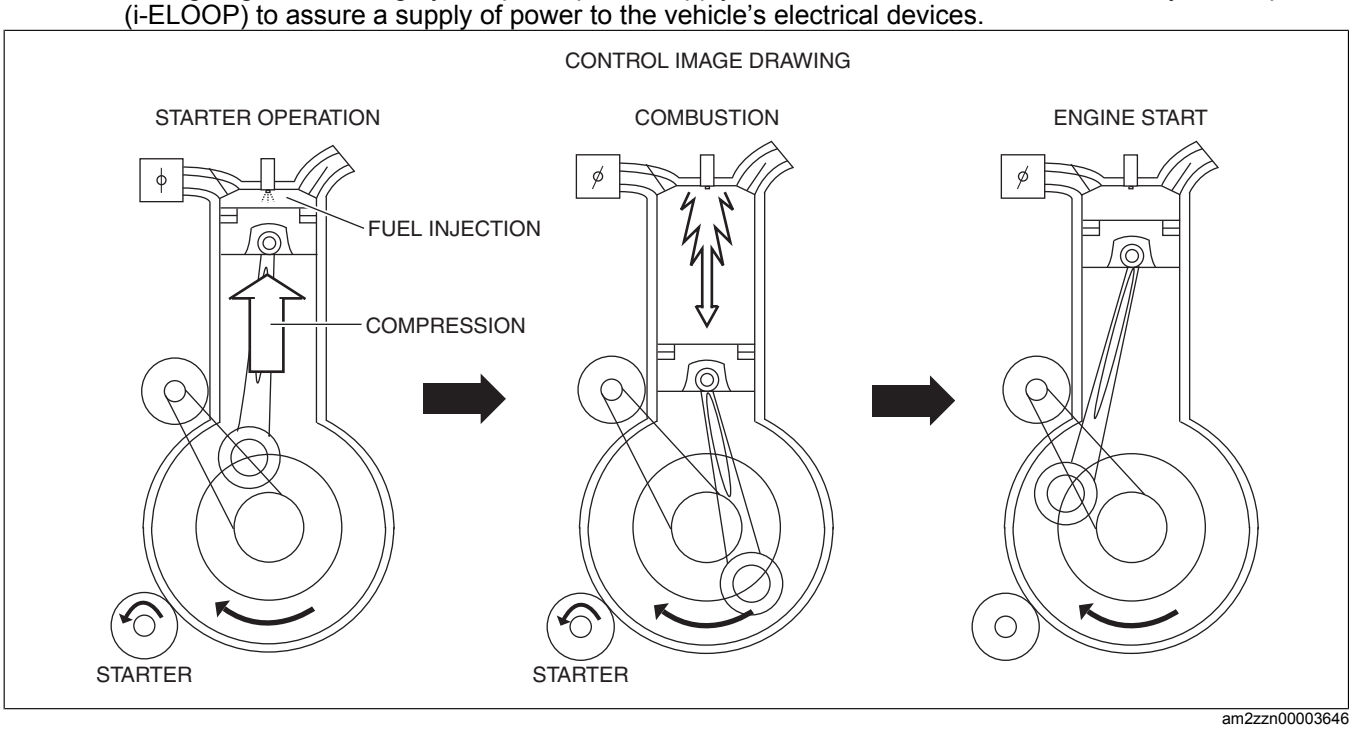
- Conditions for restarting the engine during i-stop control (engine stopped) are as follows:

Detection module	Detection item	Determination conditions		PID
		ATX	MTX	
PCM	Vehicle speed	1 km/h {0.6 mph} or more	4 km/h {2.5 mph} or more	WSPD_SEN_LF WSPD_SEN_RF WSPD_SEN_LR WSPD_SEN_RR
	Brake pedal	Brake pedal is released → depressed while in P or N position	Not applicable	TR BPA
	Brake fluid pressure	Brake fluid pressure is 0.35 MPa {3.6 kgf/cm^2, 51 psi} or less while in D or M position	Not applicable	TR BRK_F_P_C
	Accelerator pedal	Accelerator pedal is depressed while in D or M position	Not applicable	TR APP
	Clutch pedal	Not applicable	Depressed	CPP
	Steering torque	4.0 N·m {41 kgf·cm, 35 in·lbf} or more while in D or M position	Not applicable	STR_TRQ_S_M
	Steering angle	70° or more from center while in D or M position (After EPS control module learns center value)	Not applicable	STR_AB_EST
	Shift operation	Operated from P or N position to D, M, or R position	Not applicable	TR
	i-stop OFF switch	ON	←	I-Stop_OFF
	Battery charge condition	65.7% or less	←	BATT_SOC
	Power brake unit vacuum	−43 kPa {−0.44 kgf/cm^2, −6.2 psi} or more (value calculated by subtracting BARO from BBP)	←	BBP BARO
	Driver not seated (except for European (L.H.D U.K.) specs.)	• All the following conditions are met: — Seat belt (driver's side): Not fastened — Door (driver's side): Open — P or N position	• All the following conditions are met: — Seat belt (driver's side): Not fastened — Door (driver's side): Open — Neutral	SEAT_B_D DOOR_D TR CPP/PNP
	Engine stop period by i-stop	120 s or more	←	Not applicable
	Engine coolant temperature	23.5 °C {74.3 °F} or less	←	ECT
	Fuel pressure	15 MPa {153 kgf/cm^2, 2176 psi} or less	←	FRP
	System condition	An i-stop related part DTC is recorded	←	Not applicable
Start stop unit	+B (ROOM fuse) input	No	←	VPWR_B1
	Remote transmitter authorization	When engine restart by remote transmitter authorization is requested • Any door is open without remote transmitter left in vehicle • Remote transmitter is outside of vehicle after any door is opened	←	Not applicable
	System condition	An i-stop related part DTC is recorded	←	Not applicable

Detection module	Detection item	Determination conditions		PID
		ATX	MTX	
TCM	Gear position	In 2nd gear fixed mode	Not applicable	Not applicable
	TFT	20 °C {68 °F} or less, or 117°C {243 °F} or more	Not applicable	TFT
	Vehicle speed	3 km/h {2 mph} or more	Not applicable	WSPD_SEN_LF
				WSPD_SEN_RF
				WSPD_SEN_LR
				WSPD_SEN_RR
	System condition	An i-stop related part DTC is recorded	Not applicable	Not applicable
Climate control unit	A/C request (with full-auto air conditioner system)	Yes	←	STOP_ST
	A/C set temperature (with full-auto air conditioner system)	Set to MIN	←	Not applicable
	MODE dial	DEFROSTER	←	Not applicable
	System condition	An i-stop related part DTC is recorded	←	Not applicable
DSC HU/CM	System condition	An i-stop related part DTC is recorded	Not applicable	Not applicable

i-stop (engine restart control)

- When the i-stop (engine restart control) conditions are met, the PCM restarts the engine by the following controls:
  - — Fuel injection control (from first time, fuel injection to specific cylinders (expansion stroke)
    - Fuel is injected to cylinders which are determined to be stopped at the expansion stroke. During i-stop control (engine stop), cylinders are identified based on the signal from the CKP sensor and fuel injection is enabled to those cylinders which are identified first as being stopped in the expansion stroke.
  - — DC-DC converter control (without i-ELOOP)
    - When the engine is restarted by the i-stop control, the battery voltage is decreased to operate the starter by supplying power from the battery. When the starter operates, the supply voltage for electronic devices is increased by the DC-DC converter.
  - — i-ELOOP control (i-stop mode) (with i-ELOOP)
    - During engine restarting by i-stop, the power supply source is divided between the battery and capacitor (i-ELOOP) to assure a supply of power to the vehicle's electrical devices.



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Indicator illumination request

- The PCM sends an i-stop indicator light (green)/i-stop warning light (amber) illumination or flash request to the instrument cluster according to the vehicle conditions.
- The i-stop indicator light (green)/i-stop warning light (amber) illumination/flash request conditions are as follows:

IS: Engine stop by i-stop control
IR: Engine restart by i-stop control
×: Applicable
—: Not applicable

Vehicle condition			Vehicle condition	i-stop indicator light (green)/i-stop warning light (amber) display			Indicator (other)	Buzzer
				Illumination on	Flash (0.5 s)	Illumination off	Illumination on/ Flash	
Driving	IS conditions not met		—	—	—	×	—	—
	IS conditions met*6		—	Green	—	—	—	—
Vehicle parked	IS conditions not met*6		Idle	—	Green*1	—	—	—
				—	—	×	—	—
	IS conditions met*6		IS	Green	—	—	—	—
During IS	IS conditions met*6		IS	Green	—	—	—	—
	IS continues normally	**ATX** When the following conditions are met: • Brake depressed • D or M position (except 2nd gear fixed mode) • Steering angle **less than 70°** and steering torque **less than 4.0 N·m {11 kgf·cm, 9.7 in·lbf}** When the following conditions are met: • Brake released • D or M position (except 2nd gear fixed mode) →N position • D or M position (except 2nd gear fixed mode) →N position →P position	IS continues	Green	—	—	—	—
		MTX Clutch pedal depressed→ Clutch pedal is released while engine is cranking to restart by i-stop control	IS continues*9					

Vehicle condition			Vehicle condition	i-stop indicator light (green)/i-stop warning light (amber) display			Indicator (other)	Buzzer
				Illumination on	Flash (0.5 s)	Illumination off	Illumination on/ Flash	
During IS	IR request (driver's operation)	Engine start by key operation	IR	—	—	×*2	—	—
		ATX • Brake pedal released → depressed while in P or N position • Brake pedal released while in D position or M position • Brake fluid pressure is **0.35 MPa {3.6 kgf/cm^2, 51 psi} or less** in D position or M position • Accelerator pedal depressed while in D position or M position • Steering torque is **4.0 N·m {41 kgf·cm, 35 in·lbf}** or more in D or M position • Steering angle (D position or M position (except 2nd gear fixed mode)): **-70° or less or 70° or more** • Shift operation — When changed to the M position (except 2nd gear fixed mode) — P or N position →D or M or R position					
		MTX • Clutch pedal depressed						
		When the following conditions are met: • Seat belt (driver): Not fastened • Door (driver): Open • P or N position (ATX) • Neutral position (MTX)						

Vehicle condition			Vehicle condition	i-stop indicator light (green)/i-stop warning light (amber) display			Indicator (other)	Buzzer
				Illumination on	Flash (0.5 s)	Illumination off	Illumination on/ Flash	
During IS	Switches to unsafe condition (driver's operation)	Door (driver) open	IS continues	Green	—	—	—	**0.25 s** intervals
		When the following conditions are met: • Seat belt (driver): Not fastened • Door (driver): Open • D or M position (except 2nd gear fixed mode) (ATX) • In gear (MTX)	Engine stall	Amber	—	—	Same illumination as normal engine stall condition	**0.25 s** intervals for **3 s**
		Bonnet open						—
		MTX In gear	IS continues	—	Green (**0.25 s**)	—	—	—
	IR request (vehicle request)	Any of the following conditions are met: • A/C request • Battery charge **65.7% or less** • Engine stop time by the i-stop control: **120 s or more** • Engine coolant temperature: **23.5 °C {74.3 °F} or less** • Fuel pressure: 15 MPa {153 kgf/cm^2, 2176 psi} or less • System condition: An i-stop related part DTC is recorded	IR	—	—	×*2	—	—
		Any of the following conditions are met: • Vehicle speed: **1 km/h {0.6 mph} or more** (ATX) • Vehicle speed: **4 km/h {2.5 mph} or more** (MTX) • Power brake unit vacuum reduced						
		• Remote transmitter out of vehicle	IR*6	—	—	×*2	KEY warning light (red) flashing	Start stop unit normal buzzer
	IR not functional	• Cranking for **3 s or more** when engine is restarted	Engine stall	Amber	—	—	Same illumination as normal engine stall condition	—
IS permit/inhibit implement		• i-stop OFF switch on (long pressed for 0.5 s or more)	IS not authorized*7	Amber*3	—	—	—	Beep sound 1 time
		• i-stop OFF switch off (long pressed for 0.5 s or more)	IS authorized	—	—	×*4	—	Beep sound 1 time
System malfunction			IS not authorized	—	Amber*5	—	×*8	—

*1 : Changes illumination to IS when the conditions are met for driver operation (brake pedal force).

*2 : No flashing, turns off when the engine restarts.

*3 : Illuminates according to the i-stop OFF switch operation (long-press for 0.5 s or longer).

*4 : Turns off according to the i-stop OFF switch operation (long-press for 0.5 s or longer).

*5 : It may illuminate in amber when communication between PCM is cut.

*6 : The i-stop (engine-stop control) permit conditions differ depending on the driving condition. (See i-stop INDICATOR LIGHT (GREEN) [SKYACTIV-D 1.5].)

*7 : If during engine stop by i-stop control, engine restarted by i-stop control.

*8 : If there is a battery related malfunction, the master warning light illuminates.

*9 : If the clutch pedal depressed and then it is released while the engine is cranking to restart by the i-stop control, engine stop by the i-stop control continues. If the same operation is repeated several times, the engine will stall.